

Graph Aggregation Beyond Homophily Assumption

a more meaningful way to model networks

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Abstract

Data aggregation on Homophily/Heterophily networks has caused lots of discussions. Existing solutions are all based on Homophily assumption that Heterophily edges are considered as noisy data and need to be eliminated. In this paper, we propose Graph Weighted Aggregation (GWA) method and use Ricci Curvature as the force of influence between adjacent nodes. Three attributes: (1) node features, (2) network topology and (3) label information, are combined together to formulate an aggregation strategy. GWA algorithm outperforms the state-of-the-art algorithms on benchmark datasets.

CCS Concepts

• Computing methodologies → Neural networks.

Keywords

Graph Neural Networks; Homophily Assumption; Node Aggregation; Ricci Curvature

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1 Introduction

Network modeling has been discussed extensively when studying Space Networks but not Social Networks. In Social Networks, it is uncertain how feature values contribute to classification and how topological relationships influence feature-based, problem-specific patterns, how positive and negative labels are defined within the feature space.

The adage “birds of a feather flock together” succinctly captures the principle of homophily, which posits that nodes in many real-world networks tend to connect to others with similar attributes or labels [14]. Existing solutions to network aggregation primarily focus on aggregating information across varying levels of homophily. However, these methods typically rely on the homophily assumption and diminish the influence of heterophilous and random connections. In reality, connectedness and categorization represent distinct phenomena: nodes from different categories can and frequently do establish connections.

Our approach addresses the limitations of homophily-based methods. Our key contributions are summarized as follows:

- We conduct a case study on simplified networks to demonstrate the graph aggregation process.
- We introduce a dual-model approach to address graph aggregation: one model leverages Ricci curvature, the other employs Graph Neural Networks to represent node features.
- We provide theoretical proof to show Forman-Ricci curvature can be computed through graph properties.
- Through extensive experiments on eight benchmark datasets, we demonstrate that our method outperforms current state-of-the-art approaches.

2 Related Work

Existing approaches to address the homophily-heterophily issue in graph learning can be broadly categorized into three groups: node selection, weighted aggregation, and label propagation. For node selection, SparseGAD [7] categorizes neighboring nodes into homophilic, heterophilic, and irrelevant groups and utilizes only homophilic neighbors; H2GCN [28] assumes that higher homophily ratios correlate with improved performance; similarly, ASP [5] constructs different views of networks to mitigate heterophily challenges. For weighted aggregation, Personalized Propagation of Neural Predictions (PPNP) and its efficient variant APPNP [8] develop an enhanced propagation mechanism through random walk; Generalized PageRank (GPR) [6] optimizes node features using learned PageRank weights. Label Propagation (LPA) and Graph Convolutional Networks (GCN) share a conceptual similarity in propagating labels and features, respectively, via smoothing operations [26].

For curvature-enhanced network analysis, Ricci curvature has been discretized for application to networks. Ollivier-Ricci curvature [15] is computationally intensive for large-scale networks. Forman-Ricci curvature [7] serves as an abstract analog to the Bochner-Weitzenböck formula from differential geometry. Menger-Ricci and Haantjes-Ricci curvatures [23] share the topological foundation of Forman-Ricci curvature and are likewise computationally tractable for practical use.

3 Motivating Observations

We design a two-step process to test aggregation strategies on different networks. In Figure 2, when the standard deviation increases from 0.2 to 0.7, the classification performances of Linear Regression (LR) and Multi-Layer Propagation (MLP) decrease, because of the overlapping. In Figure 3, through 1-hop aggregation with addition on networks, all sample sets become the same. In Figure 4, GWA performs better than LR and MLP on different network types,



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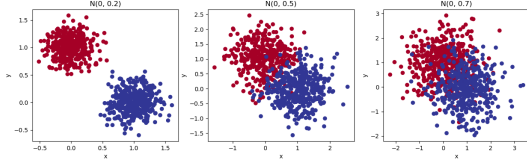


Figure 1: Visualization of Sample sets in Gaussian Distribution with Two-Class Centers at $[0, 1]$ and $[1, 0]$

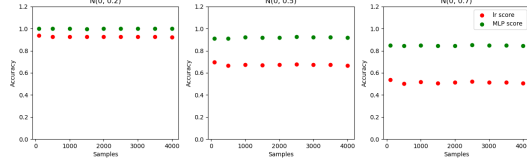


Figure 2: Classification Performance on Three Sample Sets

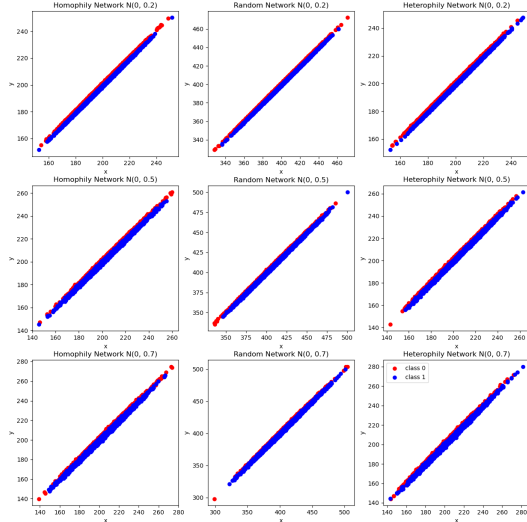


Figure 3: Visualization of Three Types of Networks after 1-hop Aggregating Neighbors on 4000 Samples

because GWA uses a different aggregation strategy introduced in following section.

- Step 1: data point generation, as shown in Figure 1, and quality evaluation, as shown in Figure 2.
- Step 2: network generation and quality evaluation.

4 Graph Weighted Aggregation (GWA)

The challenge in network modeling is the higher-order interactions in network dynamics. Different networks [3] [1] [2] encodes the dynamics and the evolution through the interactions between cells in different settings.

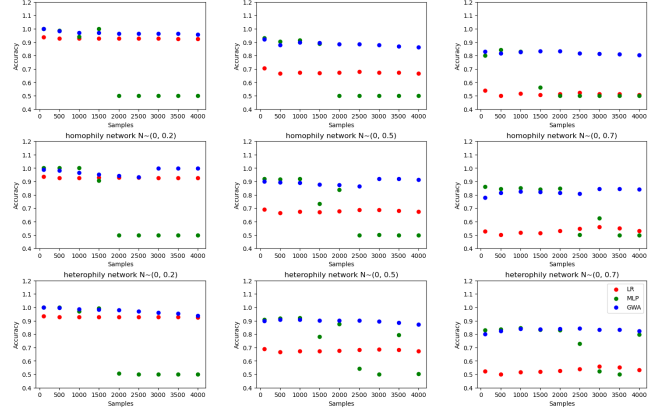


Figure 4: Visualization of Classification Performance on Different Types of Networks with Different Aggregation Methods

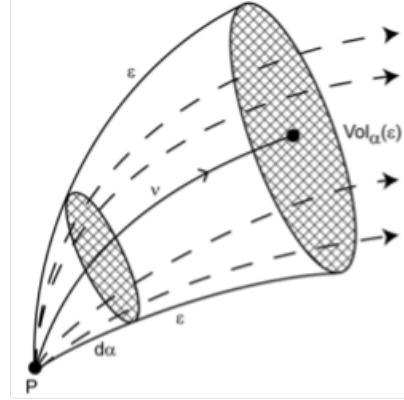


Figure 5: Geometric Interpretation of Ricci Tensor [22]

To simplify network dynamics, in Riemannian geometry, Riemannian manifold can be mapped to Tangent Space in which Ricci tensor encodes all the essential properties of a Riemannian metric [4], as shown in Figure 5, so that it can be used to compute Ricci curvature at a single point in spacetime.

To apply Riemannian geometry to networks, there are several discrete notions of Ricci curvature, such as Ollivier-Ricci curvature [15] [16] [17] [18] Forman-Ricci curvature [7], Menger-Ricci curvature [23], Haanjes- Ricci curvature [23], etc. Each captures different geometric properties and has different drawbacks because of the lacking of the smoothness in network structure [23]. In this work, we choose Forman-Ricci curvature as the foundation and use network properties to quantify Forman-Ricci curvature on both edges and nodes.

4.1 Modification of Forman-Ricci Curvature

Notations: We list the descriptions of the symbols in Table 1.

Forman-Ricci curvature in the 1-dimensional case, such as graphs or networks, can be represented in the following formula [22]:

Table 1: Descriptions of Symbols

Symbol	Description
$F(v)$	Forman-Ricci curvature for node v
w_e	Weight associated with edge e
e_{v_i}	The edge containing v_i
λ_v	The eigenvalue of node v

$$F(a) = w_a \left(\frac{w_{v_1}}{w_a} + \frac{w_{v_2}}{w_a} - \sum_{e_{v_1} \sim a, e_{v_2} \sim a} \left(\frac{w_{v_1}}{\sqrt{w_a w_{e_{v_1}}}} + \frac{w_{v_2}}{\sqrt{w_a w_{e_{v_2}}}} \right) \right) \quad (1)$$

in which a is the edge under consideration, v_1 and v_2 are two vertices, e_{v_1} and e_{v_2} are the set of edges containing v_1 and v_2 , respectively, after excluding the edge e . w_a is the weight associated with edge a , w_e is the weight associated with edge e , and w_v is the weight associated with vertex v , such as w_{v_1} is the weight for vertex v_1 and w_{v_2} is the weight for vertex v_2 .

For $w_a = w_e = w_v = 1$, $F(a) = 4 - \sum_{v \sim a} \deg(v)$.

THEOREM 1 (EDGE CURVATURE). *When graphs or networks become stabilized and reach the lowest energy, the following formula holds.*

$$F(a) = w_a (\lambda_{v_1} + \lambda_{v_2}) \quad (2)$$

in which w_a is the edge weight, v_1 and v_2 are the two ends of the edge, and λ_{v_1} and λ_{v_2} are the eigenvalues of v_1 and v_2 , respectively, when the graphs or networks are converged.

PROOF. For $w_v = \lambda_v$, in which $e \in E(G)$ and $v \in V(G)$, λ_v is the weight associated with vertex v when graphs/networks are stabilized - the eigenvalue of node v , we have

$$\begin{aligned} F(a) &= w_a * (\lambda_{v_1} + \lambda_{v_2}) - w_a * \sum_{e_{v_1} \sim a, e_{v_2} \sim a} \left(\frac{\lambda_{v_1}}{\sqrt{w_a * w_{e_{v_1}}}} + \frac{\lambda_{v_2}}{\sqrt{w_a * w_{e_{v_2}}}} \right) \\ &= w_a * (\lambda_{v_1} - \sum_{e_{v_1} \sim a} \left(\frac{\lambda_{v_1}}{\sqrt{w_a * w_{e_{v_1}}}} \right) + w_a * (\lambda_{v_2} - \sum_{e_{v_2} \sim a} \left(\frac{\lambda_{v_2}}{\sqrt{w_a * w_{e_{v_2}}}} \right)) \end{aligned}$$

Let $g_a(v) = w_a * (\lambda_{N_a(v)} - \sum_{e_{N_a(v)} \sim a} \left(\frac{\lambda_{N_a(v)}}{\sqrt{w_a * w_{e_{N_a(v)}}}} \right))$. It represents the total forces v can receive from Neighboring vertices through edge a , $N_a(v_1) = v_2$ and $N_a(v_2) = v_1$, then we have $F(a) = g_a(v_1) + g_a(v_2)$

$$\text{For } g_a(v) = w_a * (\lambda_{N_a(v)} - \sum_{e_{N_a(v)} \sim a} \left(\frac{\lambda_{N_a(v)}}{\sqrt{w_a * w_{e_{N_a(v)}}}} \right)),$$

$\sum_{e_{N_a(v)} \sim a} \left(\frac{\lambda_{N_a(v)}}{\sqrt{w_a * w_{e_{N_a(v)}}}} \right) = 0$. Because, when $w_v = \lambda_v$, adjacency matrix is stabilized so that energy propagation is not under consideration, the forces passing from higher-order vertices can be ignored.

We have $g_a(v) = w_a * \lambda_{N_a(v)}$, and then, $F(a) = w_a * \lambda_{N_a(v_2)} + w_a * \lambda_{N_a(v_1)} = w_a (\lambda_{v_1} + \lambda_{v_2})$

We can conclude $F(a) = w_a (\lambda_{v_1} + \lambda_{v_2})$ holds, when graphs or networks reach the lowest energy. $\square$

THEOREM 2 (NODE CURVATURE). *When graphs or networks become stabilized and reach the lowest energy, the following formula holds.*

$$F(v_i) = \sum_{v_j \sim N(v_i)} w_{v_i v_j} * \lambda_{v_j} \quad (3)$$

in which $w_{v_i v_j}$ is the edge weight, v_i and v_j are the two ends of the edge, and λ_{v_i} and λ_{v_j} are the eigenvalues of v_i and v_j , respectively.

PROOF. $F(v_i) = \sum_{v_j \sim N(v_i)} F(v_j)$. Based on Theorem 4.1, we have $F(v_i) = \sum_{e_{v_i}} F(e_{v_i}) - w_{e_{v_i}} * \lambda_{v_i} = \sum_{v_j \sim N(v_i)} w_{v_i v_j} * \lambda_{v_j}$

Especially, when $w_{v_i v_j} = 1$, $F(v_i) = \sum_{v_j \sim N(v_i)} \lambda_{v_j}$ $\square$

4.2 Message Passing

The classic GNN is the backbone of GWA. The momentum of the network is essential for GWA that the initial information in the network is considered as local information, and global information can be obtained by aggregating neighboring information from n hops, until the network reaches the minimum energy. During prediction, as shown in Formula 6, we disregard the structural information and only use node information as the input, because one-time aggregation on stable networks cannot cause any changes..

$$Z = M^P * X * W \quad (4)$$

$$M = C * E * A \quad (5)$$

$$Z = X * W \quad (6)$$

In Formula 4, output Z is the product of weighted adjacency matrix M after p -hop propagation, input X , and trainable weight matrix W . M is the product of node labels C , node curvatures E , and adjacency matrix A . Note that p is not allowed to be 0. However, when p is 0, GWA is turned into a one-layer MLP.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate the performance of our methodology by using eight real world datasets (as shown in Table 2) in which Cora [13], CiteSeer [9] and PubMed [24] data are citation networks, Chameleon and Squirrel data [21] are Wikipedia networks, and Cornell, Wisconsin and Texas data¹ are webpage networks.

Baselines. We compare our methodology with ten baselines: GCN [12], GAT [25], Graph-SAGE (G-S) [10], JK-Net (JK-N) [27], SSP [20], Geom-GCN (G-G) [19], GCN-LPA (G-L) [26], U-GCN (U-G) [11].

Parameter Settings. For all the methods, we use Cross Entropy loss function, Adam optimizer with learning rate 0.01 and weight decay $5e-4$, and dropout rate 0.5. We used the same splits for training, testing and validation sets. We reported 10 times' averages in Table 2.

5.2 Node Classification

We evaluate the model performance with accuracy, as shown in Table 3. On all the 8 benchmark datasets, GWA outperformed other algorithms 18.48% on Cora, 4.49% on PubMed, 34.50% on Citeseer, 37.84% on Cornell, 69.59% on Chameleon, 116.98% on Squirrel, 37.95% on Wisconsin, 37.14% on Texas. We also report how much difference MLP and GWA cause in comparison with the state-of-the-art performance, as shown in Table 4.

¹<https://www.cs.cmu.edu/afs/cs.cmu.edu/project/theo-20/www/data/>

Table 2: Data Description

Dataset	Nodes	Edges	Features	Classes
Cora	2708	5429	1433	7
PubMed	19717	44338	500	3
CiteSeer	3327	4732	3703	6
Cornell	183	298	1703	5
Wisconsin	251	515	1703	5
Texas	183	325	1703	5
Chameleon	2277	36101	2325	5
Squirrel	5201	217073	2089	5

Table 3: Model Accuracy (Percentage) on 8 Datasets (The best performance is bolded in red and the second one in black)

Alg.	Cor.	Pub.	Cit.	Cor.	Cha.	Squ.	Wis.	Tex.
GCN	82.9	83.3	73.1	46.5	52.3	33.1	47.7	52.7
GAT	83.1	84.4	72.0	48.1	51.4	32.3	46.6	49.6
SSP	81.1	79.5	71.1	55.0	21.9	19.7	49.4	55.0
JK-N	81.3	86.2	71.7	52.7	54.0	33.5	48.3	51.9
G-S	82.2	83.0	71.4	53.5	42.3	26.9	56.8	53.5
G-G	74.3	83.5	73.8	54.3	38.7	32.2	53.4	64.3
G-L	82.3	85.8	72.3	49.6	52.7	33.5	50.6	48.8
U-G	84.0	85.2	74.1	69.8	54.1	34.4	69.9	71.7
MLP	63.3	83.1	67.7	65.9	41.4	29.4	64.2	65.9
GWA	99.5	89.7	99.6	96.2	91.7	74.6	96.4	98.4

Table 4: Improvement of Benchmark and GWA Models in Comparison with the State-of-the-art Accuracy (Percentage)

Alg.	Cor.	Pub.	Cit.	Cor.	Cha.	Squ.	Wis.	Tex.
MLP	-2.1	-2.8	-6.3	-3.9	-12.7	-5.0	-5.7	-5.8
GWA	15.5	3.9	25.6	26.4	37.6	40.2	26.5	26.6

6 Limitation

This model can be applied to the networks on which node features can be influenced by the context. Some node features can not be aggregated, such as date of birth, etc.

7 Conclusion

GWA algorithm is proposed to extend Forman-Ricci curvature theory and to use graph properties to compute Forman-Ricci curvature in social networks. It is tested on simplified, synthetic networks and eight real-world benchmark networks to prove that node information, the label information and network topology together can be used to define an aggregation strategy on social networks. GWA outperforms the state-of-the-art solutions about 25.29% on average accuracy.

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